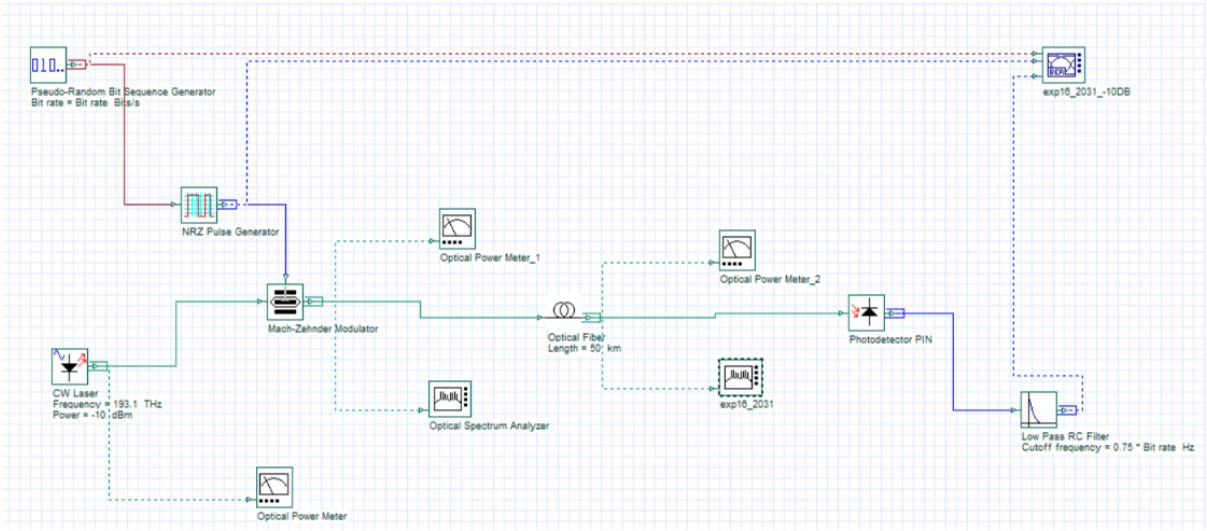
EXPERIMENT 16 : EYE DIAGRAM ANALYSIS OF OPTICAL COMMUNICATION LINK USING OPTISYSTEM

OBJECTIVES :

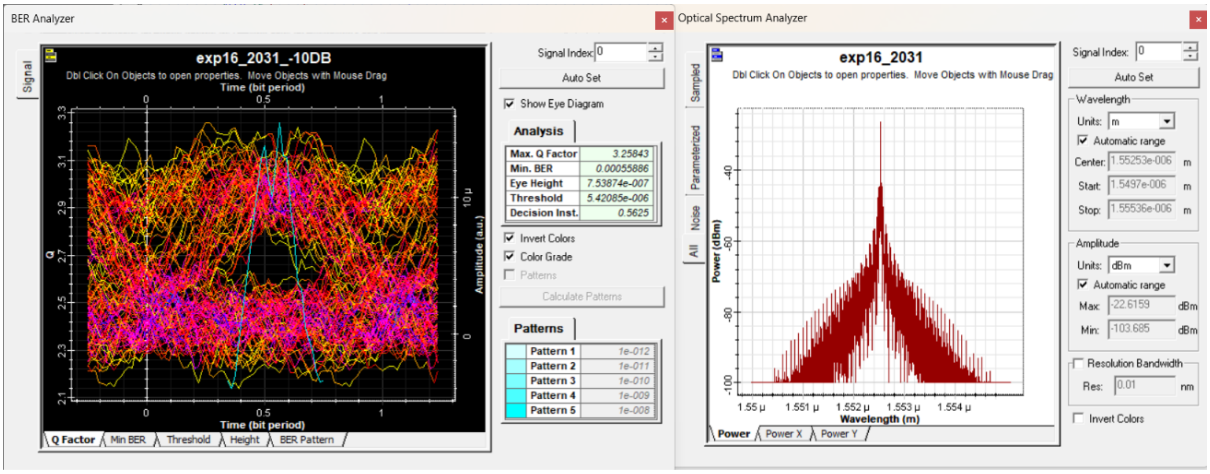
1. Design and establish an optical fiber communication link using OptiSystem software.
2. Calculate and evaluate the Bit Error Rate (BER) for different optical source power levels.
3. Analyze how variation in optical source power affects BER and overall link performance.

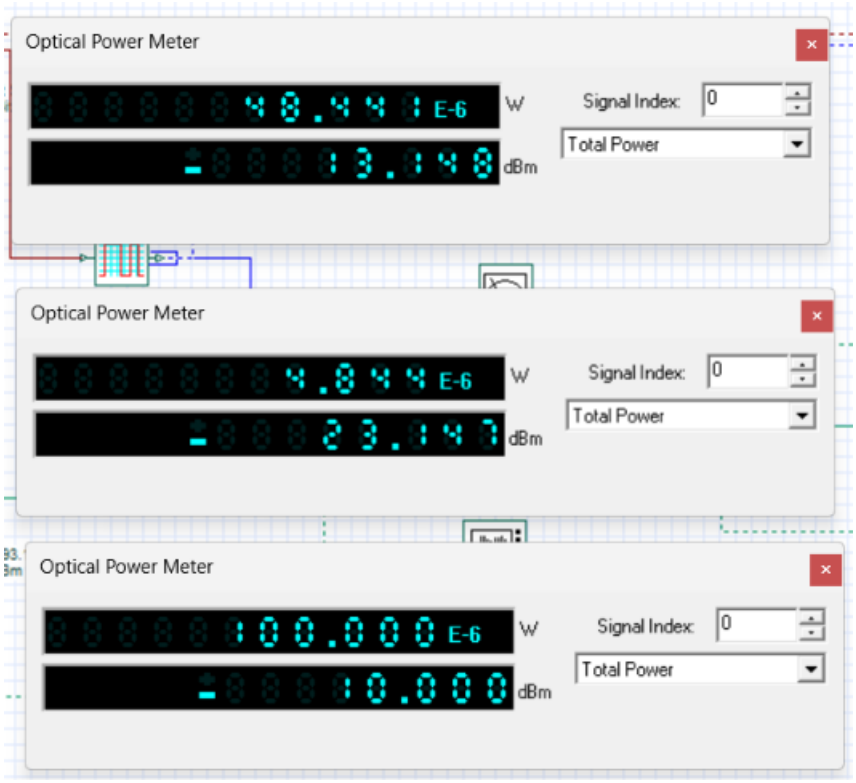
SOFTWARE USED : OPTISYSTEM SOFTWARE

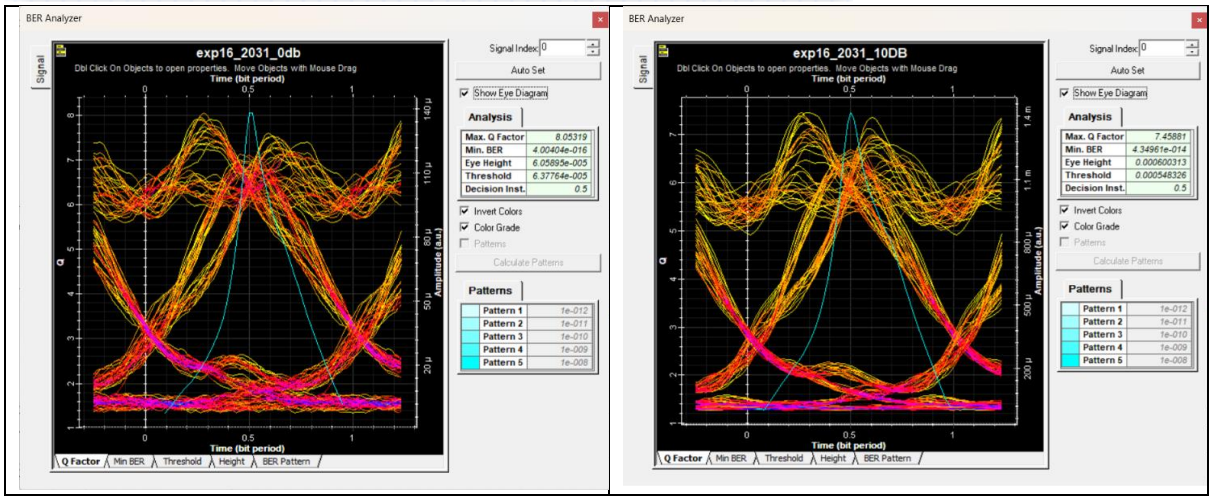
CIRCUIT DIAGRAM:



OUTPUTS:







REFLECTION:

I learned to design and simulate an optical fiber communication link using optisystem and analyze eye diagrams, BER and Q-factor at different laser power levels.

I understood how optical source power affects signal quality and reliability, identifying 0 dBm as the best-performing power level based on the measured BER and Q-factor.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & Link Setup	Correct selection and connection of OptiSystem components for the optical fiber link (PRBS Generator, NRZ Pulse Generator, CW Laser, MachZehnder Modulator, 50 km Optical Fiber, PIN Photodetector, Low Pass RC Filter, BER Analyzer).	All transmitter, modulator, fiber, receiver, and analyzer components are correctly selected, connected in sequence, and configured without errors.	All components are present with minor connection or layout errors, but the link remains functional.	Component selection is partially complete; minor connection or configuration errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20 %
Parameter Configuration & Execution	Accurate setup of input parameters: CW Laser frequency (193.1 THz), optical fiber length (50 km), and laser power levels (-10 dBm, 0 dBm, 10 dBm).	All input parameters and power-level test cases are accurately set and simulated.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but one or more power levels are missing.	Incorrect parameter settings or failure to run the required power cases.	20%
Measurement & BER Observation	Accurate extraction and recording of Max. Q-Factor, Min. BER, Eye Height, Threshold, and Decision Limit from the BER Analyzer for each laser power.	Precise data extraction for all power levels, neatly tabulated, with clear eye diagrams and BER plots captured.	Data is collected with minor omissions in Q-factor, BER, or eye parameters.	collection with Partial data incomplete tables or missing eye diagrams.	Inaccurate, incomplete, or missing data and measurements.	20 %
BER & Power Level Analysis	Detailed evaluation of the effect of optical source power on BER, Q-factor, and eye opening, including receiver sensitivity and power margin.	Comprehensive analysis correctly relating laser power to received signal strength, Q-factor, BER improvement, and saturation or nonlinear effects.	Clear analysis identifying basic trends in Q-factor and BER versus laser power.	Basic attempt at result analysis without explaining physical mechanisms such as insufficient received power or nonlinearity.	Erroneous or missing analysis of simulation results and eye diagrams.	20 %

Report & Conclusion	Identification of the optimum optical source power for reliable communication, comparison of low, moderate, and high power cases, and structural clarity.	Report features clear eye diagrams, complete output tables, and insightful conclusions identifying the optimum laser power.	Satisfactory report with relevant eye diagrams and sound concluding summary.	Incomplete report with diagrams, missing tables, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20 %

CONCLUSION:

The study shows how optical source power affects BER and signal quality in an optical communication system. Performance depends on the transmitted power level:

- **Low Power:** results in higher BER due to insufficient received signal strength.
- **Moderate Power:** gives lower BER and better signal quality.
- **High Power:** improves signal strength but may introduce nonlinear effects.

These results help in selecting the optimum optical source power for reliable and efficient optical communication